

# Explanation of Quantum Chaotic Dynamics in Matrix Gauge Theories

## PhD Thesis Committee Report

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# Outline

- 1 Introduction & Motivation
- 2 Bosonic Model
- 3 Thermodynamics
- 4 Dynamics
- 5 Entanglement
- 6 BMN Model
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# 1 Introduction & Motivation

## 2 Bosonic Model

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# Motivation: Chaos, Gravity, and the MSS Bound

- **Gauge–gravity duality** links chaotic quantum systems to black-hole physics, motivating the study of chaos in matrix models.
- **Measuring chaos:**
  - *Classical:* Lyapunov exponents  $\rightarrow$  exponential divergence of phase-space trajectories.
  - *Quantum:* Out-of-Time-Ordered Correlators (OTOCs),  
 $C(t) = \langle [\hat{W}(t), \hat{V}(0)]^2 \rangle \sim e^{2\lambda_L t}$ .
- **The MSS bound** (Maldacena–Shenker–Stanford): a universal upper bound on quantum chaos

$$\lambda_L \leq \frac{2\pi k_B T}{\hbar}.$$

Black holes are expected to *saturate* it; the SYK model does, and BFSS/BMN are conjectured to.

# Thesis Goals

## Primary Goal

Study **real-time quantum dynamics** starting from (near-)thermal-equilibrium configurations in matrix models with fuzzy-sphere geometry ( $S_F^2 \times \mathbb{R}$ ) and in the string-theoretic **BMN model**.

## Specific Objectives

- 1 Apply the **Gaussian State Approximation (GSA)** to the Heisenberg EOM  $\Rightarrow$  truncated EOM for 1- and 2-point correlators.
- 2 Determine the thermal equation of state by **von Neumann entropy maximization**.
- 3 Solve the truncated EOM numerically and extract the largest Lyapunov exponent  $\lambda_L$ .
- 4 Probe the **MSS bound**, the **Bekenstein bound**, and **entanglement dynamics**.

# What is New Since the Last Report

## Bosonic model – now *completed*

- **Non-planar interaction term** included (full  $O(2)$  model, not just planar  $D_4$ ).
- **Entropy** & Bekenstein-bound analysis.
- **Entanglement entropy** dynamics ( $\lambda_E$  vs  $\lambda_L$ ).
- Manuscript prepared for submission.

## BMN model – *ongoing*

- Full GSA EOM derived (cubic Myers term!).
- Non-trivial fuzzy-sphere vacuum  $\bar{X}_a \neq 0$ .
- $SO(3) \times SO(6)$  thermal equilibrium & gap equations.

These three pillars – non-planar term, entropy/entanglement, and the BMN model – are the focus of this talk.

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# Hamiltonian on $S_F^2 \times \mathbb{R}$

A two-matrix model on the fuzzy 2-sphere ( $i = 1, 2$ ):

$$\hat{H} = \frac{1}{2} \text{Tr} \left[ \hat{P}_i^2 + \mu^2 \hat{X}_i^2 - \frac{[L_a, \hat{X}_i]^2}{R^2} - \frac{\lambda}{2} \sum_{i \neq j} [\hat{X}_i, \hat{X}_j]^2 \right]$$

- $\hat{X}_i, \hat{P}_i$ :  $N \times N$  Hermitian matrices,  $[\hat{X}_i, \hat{P}_j] = i\delta_{ij}\mathbb{I}$ .
- $\mathcal{L}_a := [L_a, \cdot]$ : derivations on  $S_F^2$  ( $SU(2)$  algebra);  $L_a$  in spin- $\frac{N-1}{2}$  IRR.

## Two interaction potentials

- **Planar**  $\frac{\lambda}{2} \text{Tr} (\hat{X}_1 \hat{X}_1 \hat{X}_2 \hat{X}_2)$ : breaks  $O(2) \rightarrow D_4$ , analytically transparent.
- **Full**  $O(2) - \frac{\lambda}{2} \sum_{i \neq j} [\hat{X}_i, \hat{X}_j]^2$ : **planar** + **non-planar** contributions.



## Real Polarization Operator Basis

Expand fields in the real polarization operators  $Z_{lm}$  ( $l = 0, \dots, N-1$ ,  $|m| \leq l$ ):

$$\hat{X}_i = \hat{X}_i^{lm} Z_{lm}, \quad \hat{P}_i = \hat{P}_i^{lm} Z_{lm},$$

built from the standard polarization operators  $T_{lm}$ :

$$Z_{lm} = \begin{cases} \frac{1}{\sqrt{2}}(T_{lm} + T_{lm}^\dagger) & m > 0 \\ T_{l0} & m = 0 \\ \frac{i}{\sqrt{2}}(T_{lm} - T_{lm}^\dagger) & m < 0 \end{cases}$$

with  $\text{Tr}(Z_{lm} Z_{l'm'}) = \delta_{ll'} \delta_{mm'}$  and the Laplacian eigenvalues

$$\mathcal{L}_3 Z_{lm} = m Z_{lm}, \quad \mathcal{L}^2 Z_{lm} = l(l+1) Z_{lm}.$$

$\Rightarrow 2N^2$  canonical modes; in free limit,  $\omega_l^{(0)} = \sqrt{\mu^2 + l(l+1)/R^2}$ .

# Hamiltonian in the $Z_{lm}$ Basis

Taking traces yields the Hamiltonian in terms of trace coefficients:

$$\hat{H} = \frac{1}{2} \left[ (\hat{P}_i^{lm})^2 + \left( \mu^2 + \frac{l(l+1)}{R^2} \right) (\hat{X}_i^{lm})^2 \right. \\ \left. + \lambda \mathcal{H}_{m_1 m_2 m_3 m_4}^{l_1 l_2 l_3 l_4} \hat{X}_1^{l_1 m_1} \hat{X}_1^{l_2 m_2} \hat{X}_2^{l_3 m_3} \hat{X}_2^{l_4 m_4} \right]$$

$$\mathcal{H}_{m_1 m_2 m_3 m_4}^{l_1 l_2 l_3 l_4} = \mathcal{K}_{m_1 m_2 m_3 m_4}^{l_1 l_2 l_3 l_4} - \mathcal{K}_{m_1 m_3 m_2 m_4}^{l_1 l_3 l_2 l_4},$$

$$\mathcal{K} = \text{Tr} (Z_{l_1 m_1} Z_{l_2 m_2} Z_{l_3 m_3} Z_{l_4 m_4})$$

Traces  $\mathcal{K}$  reduce to Wigner  $6j$ -symbols & Clebsch–Gordan coefficients. 't Hooft limit  
 $\Rightarrow \lambda \rightarrow 1/N$ .

# Heisenberg Equations of Motion

From  $\dot{\hat{\xi}}_i^{lm} = i[\hat{H}, \hat{\xi}_i^{lm}]$ :

$$\begin{aligned}\dot{\hat{X}}_1^{lm} &= \hat{P}_1^{lm}, & \dot{\hat{X}}_2^{lm} &= \hat{P}_2^{lm}, \\ \dot{\hat{P}}_1^{lm} &= -\left[\left(\mu^2 + \frac{l(l+1)}{R^2}\right)\hat{X}_1^{lm} + \frac{\lambda}{2}\mathcal{N}_{m\,m_1m_2m_3}^{l\,l_1l_2l_3}\hat{X}_1^{l_1m_1}\hat{X}_2^{l_2m_2}\hat{X}_2^{l_3m_3}\right], \\ \dot{\hat{P}}_2^{lm} &= -\left[\left(\mu^2 + \frac{l(l+1)}{R^2}\right)\hat{X}_2^{lm} + \frac{\lambda}{2}\mathcal{N}_{m\,m_1m_2m_3}^{l\,l_1l_2l_3}\hat{X}_2^{l_1m_1}\hat{X}_1^{l_2m_2}\hat{X}_1^{l_3m_3}\right].\end{aligned}$$

↓

$$\mathcal{N}_{m_1m_2m_3m_4}^{l_1l_2l_3l_4} = \mathcal{K}_{m_1m_2m_3m_4}^{l_1l_2l_3l_4} + \mathcal{K}_{m_2m_1m_3m_4}^{l_2l_1l_3l_4} - \mathcal{K}_{m_1m_3m_2m_4}^{l_1l_3l_2l_4} - \mathcal{K}_{m_2m_3m_1m_4}^{l_2l_3l_1l_4}$$

$4N^2$  coupled nonlinear ODEs – intractable in general. We need an approximation that captures quantum effects.

# Gaussian State Approximation (GSA)

Represent the state by the most general **Gaussian Wigner function**:

$$\mathcal{W}_{\hat{\rho}}(\xi) = \mathcal{N} e^{-\frac{1}{2}(\xi - \bar{\xi})^T \Sigma^{-1} (\xi - \bar{\xi})}, \quad \bar{\xi}_a = \langle \hat{\xi}_a \rangle,$$

fully characterized by the 1-point functions  $\bar{\xi}$  and the covariance matrix

$$\Sigma_{ab} = \langle \langle \hat{\xi}_a \hat{\xi}_b \rangle \rangle = \frac{1}{2} \langle \hat{\xi}_a \hat{\xi}_b + \hat{\xi}_b \hat{\xi}_a \rangle - \langle \hat{\xi}_a \rangle \langle \hat{\xi}_b \rangle.$$

## Key mechanism – Wick's theorem

Higher correlators factorize into 1- and 2-point functions:

$$\langle \delta \hat{\xi}_{a_1} \cdots \delta \hat{\xi}_{a_{2n}} \rangle_s = \sum_{\text{pairings}} \prod \langle \langle \hat{\xi}_{a_i} \hat{\xi}_{a_j} \rangle \rangle, \quad \langle \text{odd} \rangle = 0.$$

⇒ The infinite Schwinger–Dyson hierarchy *truncates* to a closed set of nonlinear ODEs for  $\langle X \rangle, \langle P \rangle, \langle \langle XX \rangle \rangle, \langle \langle XP \rangle \rangle, \langle \langle PP \rangle \rangle$ .

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# Thermal States from Symmetry

The thermal state inherits the symmetries of  $\hat{H}$ :

- **Parity & time reversal**  $\Rightarrow \langle \hat{X}_i^{lm} \rangle = \langle \hat{P}_i^{lm} \rangle = 0$ ,  $\langle \langle X_i^{lm} P_j^{l'm'} \rangle \rangle = 0$ .
- **$O(2)$  (or  $D_4$ )**  $\Rightarrow \langle \langle X_i X_j \rangle \rangle \sim \delta_{ij}$ .
- **$SU(2)$**   $\Rightarrow$  correlators depend only on  $l$ :

$$\langle \langle X_i^{lm} X_j^{l'm'} \rangle \rangle = \sigma_{xx}(l) \delta_{ij} \delta^{ll'} \delta^{mm'}, \quad \langle \langle P_i^{lm} P_j^{l'm'} \rangle \rangle = \sigma_{pp}(l) \delta_{ij} \delta^{ll'} \delta^{mm'}.$$

## Symplectic eigenvalues

The matrix  $\Sigma\Omega$  has eigenvalues  $\pm i f_l$  ( $2l + 1$ -fold degenerate), with

$$f_l = \sqrt{\sigma_{xx}(l) \sigma_{pp}(l)}.$$

Only  $N$  independent eigenvalues. Pure state  $\Leftrightarrow f_l = \frac{1}{2} \forall l$ .

# Entropy and Energy

NEW

By Williamson's theorem the **von Neumann entropy** is a function of the  $f_l$ :

$$S = 2 \sum_{l=0}^{N-1} (2l+1) \left[ \left(f_l + \frac{1}{2}\right) \ln \left(f_l + \frac{1}{2}\right) - \left(f_l - \frac{1}{2}\right) \ln \left(f_l - \frac{1}{2}\right) \right].$$

The energy  $E = \langle \hat{H} \rangle$  in the thermal state:

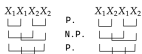
$$E = \sum_l (2l+1) \left( \sigma_{pp}(l) + \left(\mu^2 + \frac{l(l+1)}{R^2}\right) \sigma_{xx}(l) \right) + \frac{\lambda C^2}{2N},$$

where the **coordinate dispersion** plays a central role:

$$C := \sum_{l=0}^{N-1} (2l+1) \sigma_{xx}(l) = \frac{1}{2} \langle \langle \text{Tr} (X_i X_i) \rangle \rangle, \quad \frac{1}{N} \langle \langle \text{Tr} X_i X_i \rangle \rangle = \frac{2C}{N}.$$

## Planar vs. Non-Planar: The Wick Contractions

NEW



Planar (P.) and Non-Planar (N.P.) contractions of the quartic interaction.

The interaction

$\frac{\lambda}{2} (\text{Tr } X_1 X_1 X_2 X_2 - \text{Tr } X_1 X_2 X_1 X_2)$  gives two channels:

- **Planar:** clean factorization

$$\mathcal{K}^{l_1 l_1 l_2 l_2} \sigma_{xx}(l_1) \sigma_{xx}(l_2) = \frac{1}{N} C^2.$$

- **Non-planar:** brings in the  $6j$ -symbol

$$G(l_1, l_2) = (-1)^{\dots} \begin{Bmatrix} l_1 & \frac{N-1}{2} & \frac{N-1}{2} \\ l_2 & \frac{N-1}{2} & \frac{N-1}{2} \end{Bmatrix}.$$

**Crucial point:** because the symmetry is  $SU(2)$  (not  $SU(N)$ ), the non-planar term is **not**  $1/N^2$ -suppressed!



## Equation of State: Self-Consistency

Entropy maximization at fixed energy ( $\mathcal{L} = S - \beta(E - E_{eq})$ ) gives the thermal ratio

$$\frac{\sigma_{pp}(l)}{\sigma_{xx}(l)} = \omega_l^2, \quad f_l = \frac{1}{2} \coth\left(\frac{\beta\omega_l}{2}\right), \quad \beta = 1/T.$$

### Planar

Single self-consistency equation for  $C$ :

$$\omega_l = \sqrt{\mu^2 + \frac{l(l+1)}{R^2} + \frac{\lambda C}{N}}.$$

### Full (non-planar)

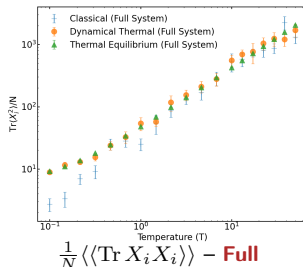
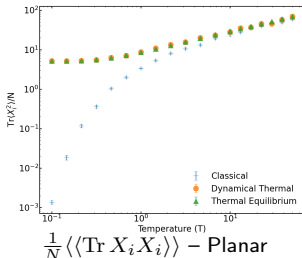
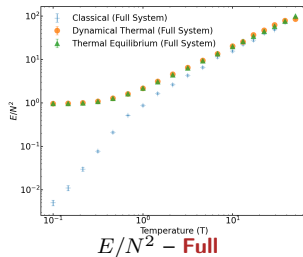
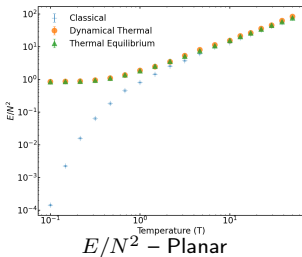
$N$  coupled equations via  
 $D(l) = \sum_{l'} (2l' + 1) \sigma_{xx}(l') G(l, l')$ :

$$\omega_l = \sqrt{\mu^2 + \frac{l(l+1)}{R^2} + \frac{\lambda C}{N} - \lambda D(l)}.$$

$$\sigma_{xx}(l) = \frac{1}{2\omega_l[\vec{\sigma}]} \coth\left(\frac{\beta\omega_l[\vec{\sigma}]}{2}\right), \quad l = 0, \dots, N-1 \quad (\text{solved numerically}).$$

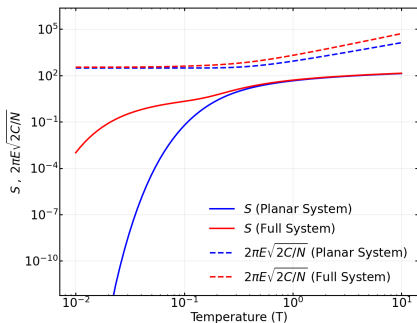
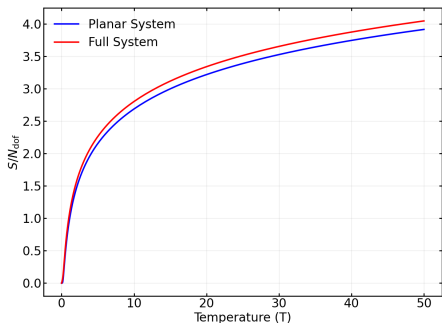
# Energy & Coordinate Dispersion vs. Temperature

NEW



# Entropy & the Bekenstein Bound

NEW



$S/N_{\text{dof}}$  vs  $T$  ( $N_{\text{dof}} = 2N^2 = 50$ ).

$S$  vs the bound  $2\pi E\sqrt{2C/N}$ .

- Entropy increases monotonically with  $T$ ; the small dent (full, red) near  $T \approx 0.1$  is the decoupled  $l = 0$  mode.
- Bekenstein bound**  $S \leq 2\pi EK$  with  $K = \sqrt{\frac{1}{N} \langle \langle \text{Tr } X_i X_i \rangle \rangle}$  is comfortably satisfied:

$$S \ll 2\pi E\sqrt{2C/N}.$$

# Choosing Dynamical Initial Conditions

## The problem

The thermal state is a *static* solution of the EOM  $\Rightarrow$  no chaos from small fluctuations.

**Solution:** split each dispersion into quantum + classical parts,

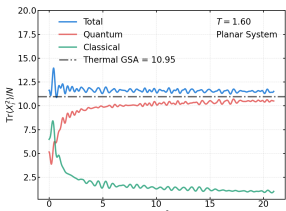
$$\sigma_{\xi\xi}(l) = \sigma_{\xi\xi}^0(l) + \sigma_{\xi\xi}^c(l),$$

- $\sigma^0$ : zero-temperature (pure-state) quantum dispersion.
- $\sigma^c$ : classical thermal dispersion (difference from  $T = 0$ ).

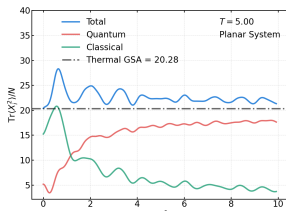
Build a classical mixture of **shifted pure Gaussian states**  $|\bar{X}_i^{lm}, \bar{P}_i^{lm}\rangle$  with 1-point functions drawn randomly from Gaussians of variance  $\sigma^c$ . This **dynamical thermal state** is time-dependent, sits on the thermal equation of state, and encodes the chaotic motion.

# Dynamical Thermal States Track Equilibrium

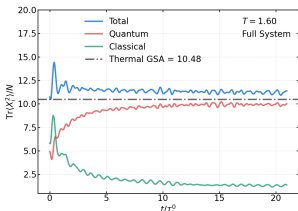
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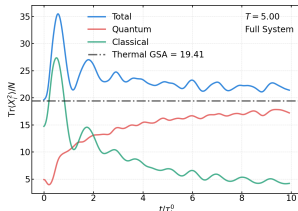
Planar,  $T = 1.6$



Planar,  $T = 5$



Full,  $T = 1.6$



Full,  $T = 5$

$\frac{1}{N} \langle \langle \text{Tr } XX \rangle \rangle$  vs time (units of  $\tau_L = \lambda_L^{-1}$ ). It fluctuates with small amplitude just above its equilibrium value.

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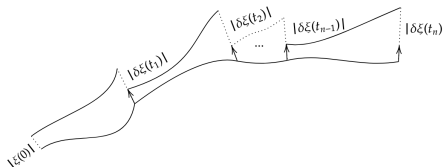
## ④ Dynamics

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# Computing the Largest Lyapunov Exponent



Renormalize-and-sum procedure for  $\lambda_L$ .

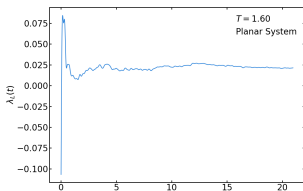
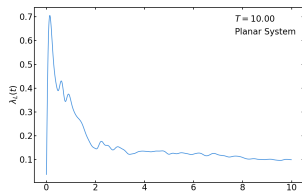
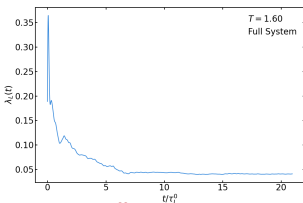
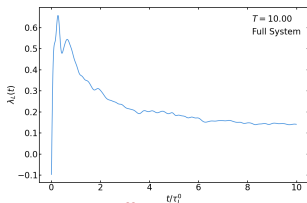
$$\lambda_L = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{\|\delta\xi(t)\|}{\|\delta\xi(0)\|} \longrightarrow \lambda_L \approx \frac{1}{n\Delta t} \sum_{i=1}^n \ln \frac{\|\delta\xi(t_i)\|}{\|\delta\xi(0)\|}.$$

**Implementation:** C++ code, **symplectic Euler** integrator (update order  $P \rightarrow XP \rightarrow PP \rightarrow XX \rightarrow X$ ), run on the TRUBA HPC cluster with SLURM job arrays.

- Integrate two nearby trajectories,  $\|\delta\xi(0)\| \approx 10^{-5}$ .
- Measure separation, renormalize, accumulate growth.
- Average over 7–9 random initial conditions.

$\lambda_L$  Time Series: Transient  $\rightarrow$  Saturation

NEW

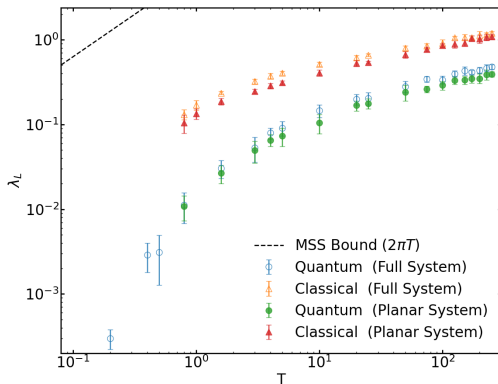
Planar,  $T = 1.6$ Planar,  $T = 10$ Full,  $T = 1.6$ Full,  $T = 10$ 

After a transient within the first  $\sim 2\tau_L^{(0)}$  (planar) /  $\sim 6\tau_L^{(0)}$  (full),  $\lambda_L$  converges to a stable saturation value.



Main Result:  $\lambda_L$  vs. Temperature & the MSS Bound

NEW

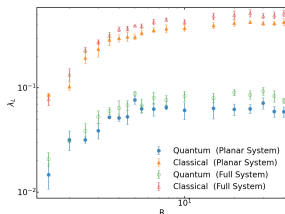
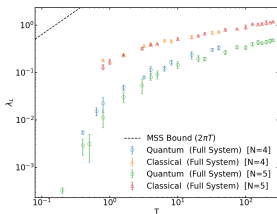
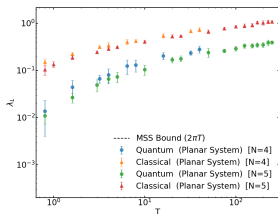


- **MSS bound  $\lambda_L \leq 2\pi T$  respected at all  $T$ .**
- $\lambda_L \rightarrow 0$  at finite  $T_c$ :  
**quantum effects suppress chaos** as  $T$  drops.
  - Planar:  $T_c \lesssim 0.8$
  - Full:  $T_c \approx 0.1$
- Quantum  $\rightarrow$  classical as  $T \gtrsim 10^2$ , always staying below.
- Full system slightly more chaotic than planar everywhere.

$N = 5$ ,  $R = 5$ ,  $\mu = 1/10$ ,  $\lambda = 1/5$ . This plot is the central physics result of the completed bosonic work.

# $N$ -dependence and Curvature ( $R$ ) dependence

NEW

Planar:  $N = 4$  vs  $N = 5$ .Full:  $N = 4$  vs  $N = 5$ . $\lambda_L$  vs radius  $R$  at  $T = 5$ .

- **$N$ -scaling:**  $\lambda_L$  scales alike at  $N = 4$  and  $N = 5$  for both interactions.
- **Curvature:**  $\lambda_L$  is steady for  $R \gtrsim 3$ , decreasing slowly for smaller  $R$  – the  $l(l+1)/R^2$  term acts as an effective mass, driving the system more harmonic (less chaotic).

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# Entanglement Entropy via Reduced Covariance

NEW

## Why this is feasible

GSA is not unitary, but **pure states stay pure**  $\Rightarrow$  entanglement entropy is computable.

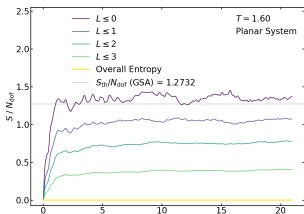
Sequence of bipartitions of the  $2N^2$  modes by angular momentum:

$$\mathcal{H} = \mathcal{H}_{\leq L} \otimes \mathcal{H}_{>L}, \quad \mathcal{H}_{\leq L} : \text{modes with } l \leq L.$$

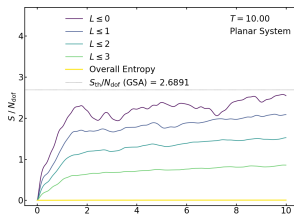
- Reduced covariance  $\Sigma_{\leq L}$ : restrict  $\Sigma$  to the first  $(L+1)^2$  modes.
- $S_{\leq L}$  = von Neumann entropy from symplectic eigenvalues of  $\Omega_{\leq L} \Sigma_{\leq L}(t)$ .
- The **fuzzy structure** naturally plays the role of a lattice discretization, while keeping  $SU(2)$  intact – and the method applies directly to the *interacting* theory.

# Entanglement: Fast-Scrambling Behavior

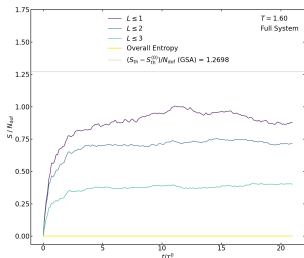
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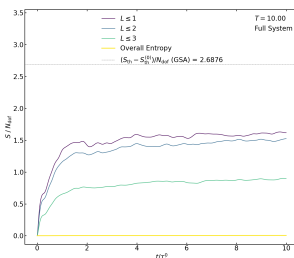
Planar,  $T = 1.6$



Planar,  $T = 10$



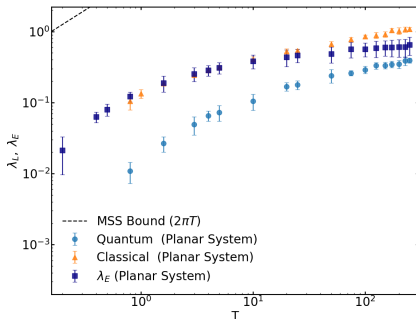
Full,  $T = 1.6$



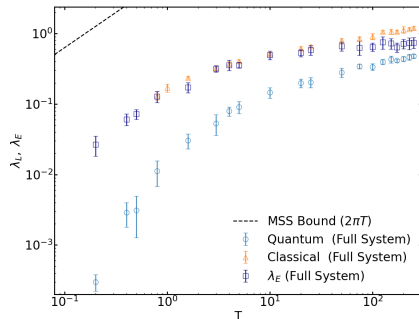
Full,  $T = 10$

# Entanglement Saturation vs. Lyapunov: $\lambda_E$ vs $\lambda_L$

NEW



Planar System.



Full System.

Fitting  $S_{\leq L} \sim \tilde{S} \tanh(t/\tau_E)$  defines  $\lambda_E := \tau_E^{-1}$ :

- $\lambda_E > \lambda_L$  **at all temperatures** – entanglement saturates faster than trajectories diverge.
- $\lambda_E$  tracks the *classical*  $\lambda_L$ ; the gap widens as the system gets colder.

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# The BMN Matrix Model: Why It Is the Natural Next Step

NEW

## What is the BMN model?

A **mass deformation of BFSS** that maximally preserves supersymmetry. Its classical vacua are **fuzzy 2-spheres** – exactly the geometry of our completed work.

- Global symmetry  $SO(3) \times SO(6)$  (vs.  $SO(2)$  in the bosonic model).
- Nine  $N \times N$  Hermitian matrices  $\hat{X}_I$ , split as  $a \in \{1, 2, 3\}$  ( $SO(3)$ ),  $\alpha \in \{4, \dots, 9\}$  ( $SO(6)$ ).
- Holographically related to black holes  $\Rightarrow$  the prime target for testing the MSS bound.

We have completed the full GSA derivation and the thermal-equilibrium analysis.  
Numerical work is underway.



# The BMN Hamiltonian

NEW

$$\hat{H} = \frac{1}{2} \text{Tr} \left( \hat{P}_I^2 + \sum_{I=1}^9 \Omega_I^2 \hat{X}_I^2 - \frac{\lambda}{2} \sum_{I \neq J} [\hat{X}_I, \hat{X}_J]^2 + \frac{2i\mu\sqrt{\lambda}}{3} \epsilon_{abc} \hat{X}_a \hat{X}_b \hat{X}_c \right)$$

**Cubic term:** the new structural ingredient absent from the bosonic model.

with  $\Omega_I^2$  defined as

$$\Omega_I^2 = \begin{cases} \mu^2/9 & I \in \{1, 2, 3\} \\ \mu^2/36 & I \in \{4, \dots, 9\} \end{cases}$$

- The cubic term introduces the structure constant

$$\mathcal{F}_{m_1 m_2 m_3}^{l_1 l_2 l_3} = \text{Tr} (Z_{l_1 m_1} Z_{l_2 m_2} Z_{l_3 m_3}).$$

- Symmetrized:  $\tilde{\mathcal{F}}_{m_1 m_2 m_3}^{l_1 l_2 l_3} = 2\mathcal{F}_{m_1 m_2 m_3}^{l_1 l_2 l_3} - \mathcal{F}_{m_1 m_3 m_2}^{l_1 l_3 l_2}.$

## BMN: Heisenberg &amp; One-Point EOM

NEW

Heisenberg EOM ( $\partial_t \hat{\mathcal{O}} = -i[\hat{\mathcal{O}}, \hat{H}]$ ) for the momentum:

$$\dot{\hat{P}}_K^{l_k m_k} = -\left[\Omega_K^2 \hat{X}_K^{l_k m_k} + i\frac{\mu\sqrt{\lambda}}{3}\epsilon_{Kbc}\tilde{\mathcal{F}}\hat{X}_b\hat{X}_c - \frac{\lambda}{2}\mathcal{N}\hat{X}_J\hat{X}_K\hat{X}_J\right].$$

## GSA: mean-field + fluctuation decomposition

Write  $\hat{X} = \bar{X} + \delta\hat{X}$  with  $\bar{X}_I^{lm} = \langle\hat{X}_I^{lm}\rangle$ ,  $\langle\delta\hat{X}\rangle = 0$ . Then the one-point  $\langle\hat{P}\rangle$  EOM closes:

$$\begin{aligned}\dot{\bar{P}}_K^{l_k m_k} = & -\left[\Omega_K^2 \bar{X}_K^{l_k m_k} + i\frac{\mu\sqrt{\lambda}}{3}\epsilon_{Kbc}\tilde{\mathcal{F}}(\bar{X}_b\bar{X}_c + \langle\langle X_b X_c \rangle\rangle)\right. \\ & \left. - \frac{\lambda}{2}\mathcal{N}(\bar{X}_J\bar{X}_K\bar{X}_J + \langle\langle X_J X_K \rangle\rangle\bar{X}_J + \dots)\right].\end{aligned}$$

Each classical product is dressed by its 2-point (quantum) correction – the GSA backreaction.

## BMN: Two-Point EOM

NEW

Differentiating the connected covariances yields the kinematic relations

$$\partial_t \langle \langle X_I X_J \rangle \rangle = \langle \langle P_I X_J \rangle \rangle + \langle \langle X_I P_J \rangle \rangle,$$

$$\partial_t \langle \langle X_I P_J \rangle \rangle = \langle \langle P_I P_J \rangle \rangle + \langle \langle X_I \dot{P}_J \rangle \rangle,$$

$$\partial_t \langle \langle P_I P_J \rangle \rangle = \langle \langle \dot{P}_I P_J \rangle \rangle + \langle \langle P_I \dot{P}_J \rangle \rangle.$$

The force-correlators  $\langle \langle X_I \dot{P}_J \rangle \rangle$ ,  $\langle \langle P_I \dot{P}_J \rangle \rangle$  are expanded term by term via Wick's theorem. The quartic piece contributes three pairing channels, e.g.

$$\begin{aligned} \langle \langle X_I \dot{P}_J \rangle \rangle_{\text{quart}} = \frac{\lambda}{2} \mathcal{N} \Big[ & \langle \langle X_I X_K \rangle \rangle \bar{X}_J \bar{X}_K + \bar{X}_K \langle \langle X_I X_J \rangle \rangle \bar{X}_K + \dots \\ & + \langle \langle X_I X_K \rangle \rangle \langle \langle X_J X_K \rangle \rangle + \langle \langle X_I X_J \rangle \rangle \langle \langle X_K X_K \rangle \rangle + \dots \Big]. \end{aligned}$$

$\Rightarrow$  The full closed truncated system for the BMN model has been derived (mass + cubic + quartic).

# BMN Thermal Equilibrium: Non-Trivial Vacuum

NEW

## Key difference from the bosonic model

The BMN classical vacuum is **non-zero**:  $\bar{X}_a^{\text{vac}} = \frac{\mu}{3\sqrt{\lambda}} L_a$ ,  $\bar{X}_\alpha^{\text{vac}} = 0$ .

Converting  $L_a$  to the real polarization basis ( $l = 1$  sector only) gives the thermal one-point function

$$\bar{X}_a^{lm} = \alpha \delta_{l,1} \Lambda_a^m, \quad \alpha = \frac{\mu}{3\sqrt{\lambda}} \sqrt{\frac{S(S+1)(2S+1)}{3}}, \quad S = \frac{N-1}{2}.$$

The unbroken symmetries reduce the covariance to a block-isotropic form with **two** species sectors:

$$\langle\langle X_I^{l_1 m_1} X_J^{l_2 m_2} \rangle\rangle = \sigma_{xx}^{(I)}(l_1) \delta_{IJ} \delta^{l_1 l_2} \delta^{m_1 m_2}, \quad \sigma^{(I)} = \begin{cases} \sigma^{(3)} & I \leq 3 \\ \sigma^{(6)} & I \geq 4 \end{cases}$$

# BMN: Energy Functional & Gap Equations

NEW

The Gaussian energy functional (classical background drops out by stationarity):

$$E[\vec{\sigma}] = \frac{1}{2} \sum_l (2l+1) [3\sigma_{pp}^{(3)} + 6\sigma_{pp}^{(6)}] + \frac{\mu^2}{6} C_3 + \frac{\mu^2}{12} C_6 \\ + \frac{\lambda}{2N} [6C_3^2 + 36C_3C_6 + 30C_6^2] + \dots$$

Entropy maximization  $\Rightarrow$  **gap equations** for the two frequency sectors:

$$(\omega_l^{(3)})^2 = \frac{\mu^2}{9} + \frac{2\lambda}{N} (2C_3^{\text{tot}} + 6C_6) - 2\lambda [2D^{(3),\text{tot}}(l) + 6D^{(6)}(l)], \\ (\omega_l^{(6)})^2 = \frac{\mu^2}{36} + \frac{2\lambda}{N} (3C_3^{\text{tot}} + 5C_6) - 2\lambda [3D^{(3),\text{tot}}(l) + 5D^{(6)}(l)].$$

closed by the **coupled  $2N$  self-consistency system**

$$\sigma_{xx}^{(s)}(l) = \frac{1}{2\omega_l^{(s)}[\vec{\sigma}]} \coth\left(\frac{\beta\omega_l^{(s)}[\vec{\sigma}]}{2}\right), \quad s \in \{3, 6\}.$$

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## Summary

### Completed – Bosonic model on $S_F^2 \times \mathbb{R}$

- Full GSA truncated EOM, thermal equation of state via entropy maximization.
- **Non-planar term included** – not  $1/N^2$ -suppressed (only  $SU(2)$  symmetry).
- $\lambda_L$  **respects the MSS bound** at all  $T$ ; vanishes at finite  $T_c$ .
- **Bekenstein bound satisfied**; **entanglement** shows fast scrambling with  $\lambda_E > \lambda_L$ .
- Manuscript prepared for submission.

### Ongoing – BMN model

- Full GSA EOMs and  $SO(3) \times SO(6)$  thermal equilibrium **derived**.
- Gap equations & coupled self-consistency system in hand.

# Timeline & Outlook

| Task                                             | Status      | Target    |
|--------------------------------------------------|-------------|-----------|
| Bosonic: GSA EOM + thermal equilibrium           | Complete    | —         |
| Bosonic: Lyapunov + entanglement                 | Complete    | —         |
| Bosonic: Manuscript                              | Complete    | Submitted |
| BMN: GSA EOM derivation                          | Complete    | —         |
| BMN: Thermal equilibrium derivation              | Complete    | —         |
| BMN: Self-consistency solver                     | In progress | Q2 2026   |
| BMN: Lyapunov computation (C++)                  | In progress | Q2 2026   |
| BMN: Entanglement entropy analysis               | Planned     | Q2 2026   |
| BMN: Manuscript                                  | Planned     | Q3 2026   |
| Parallel Project: TFD & Bogoliubov tranformation | Planned     | Q4 2026   |
| Thesis finalization                              | Planned     | May 2027  |



**Thank you!**

Questions & Discussion



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